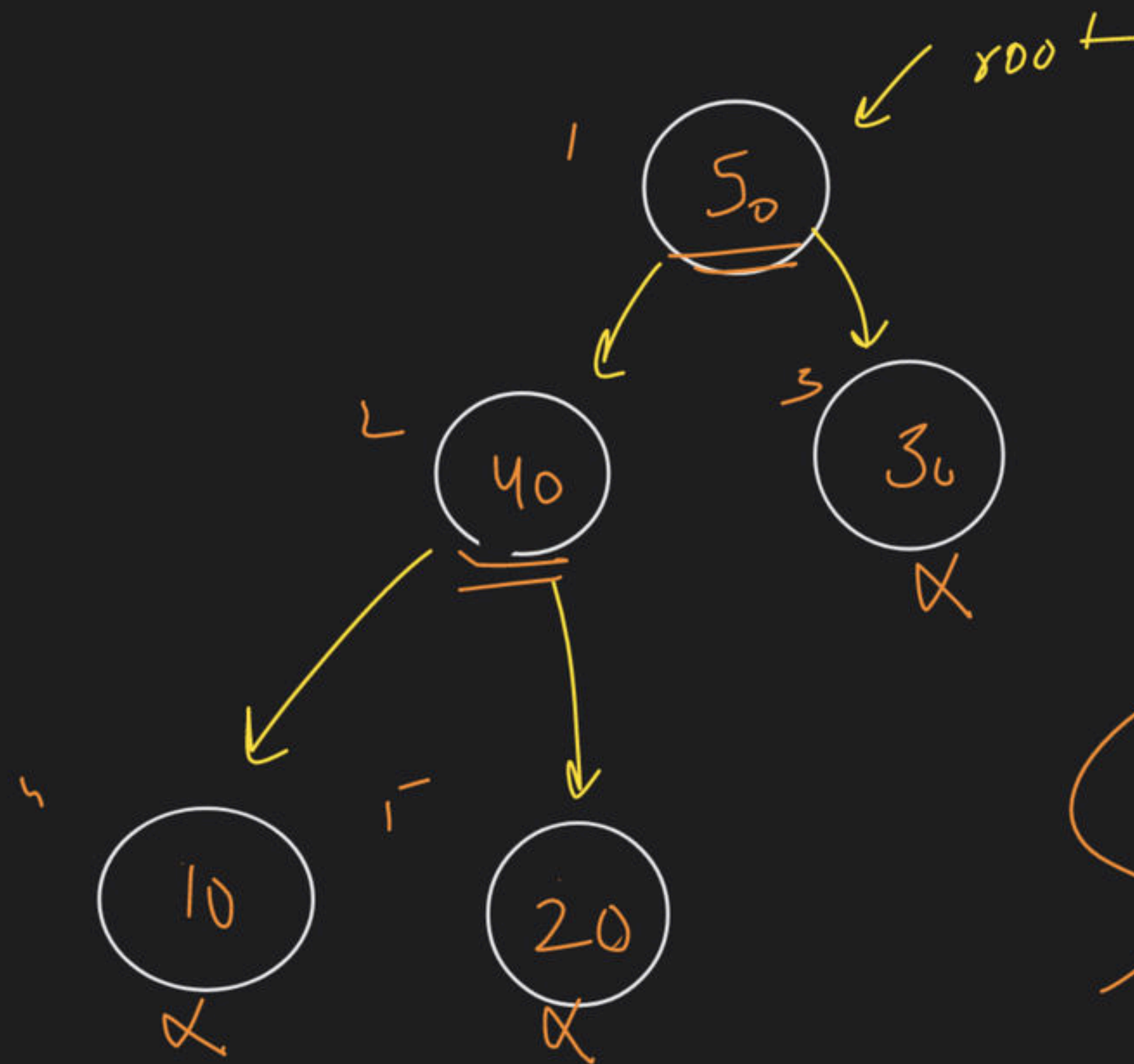


Heaps Class - 2

Special class



50 40 30 10 20

Heap

9

D.S.

C.B.T + Heap Property Satisfy

Min / Max

parent / root

parent / root

$> l, > r$

$\langle l \rangle, \langle v \rangle$

$C \cdot B \cdot I$

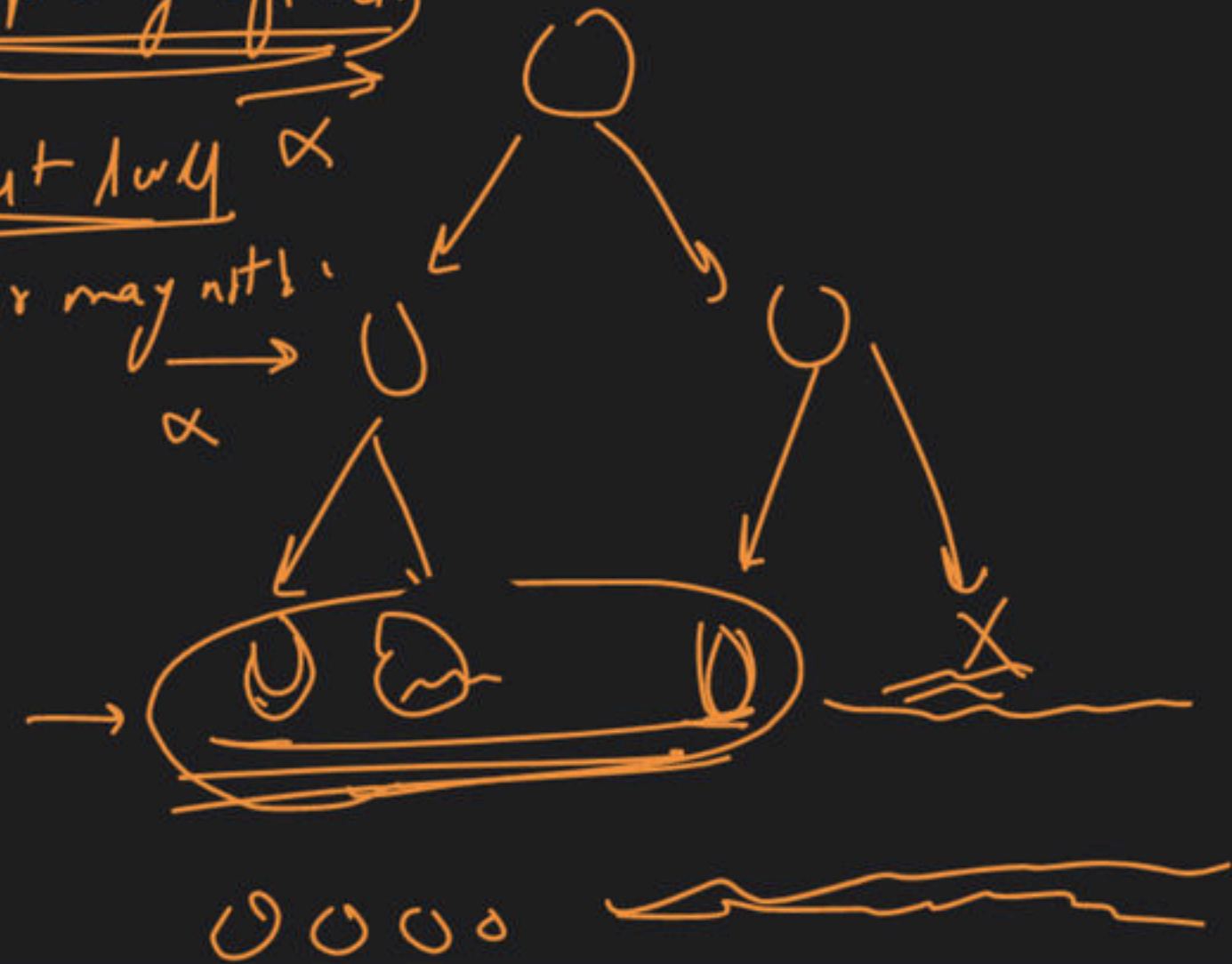
except last level - α

all levels are completely filled

→ ex (op) the last word $\times$
 may or may not be $\rightarrow$

② nodes fill

left → right. [continuous manner]



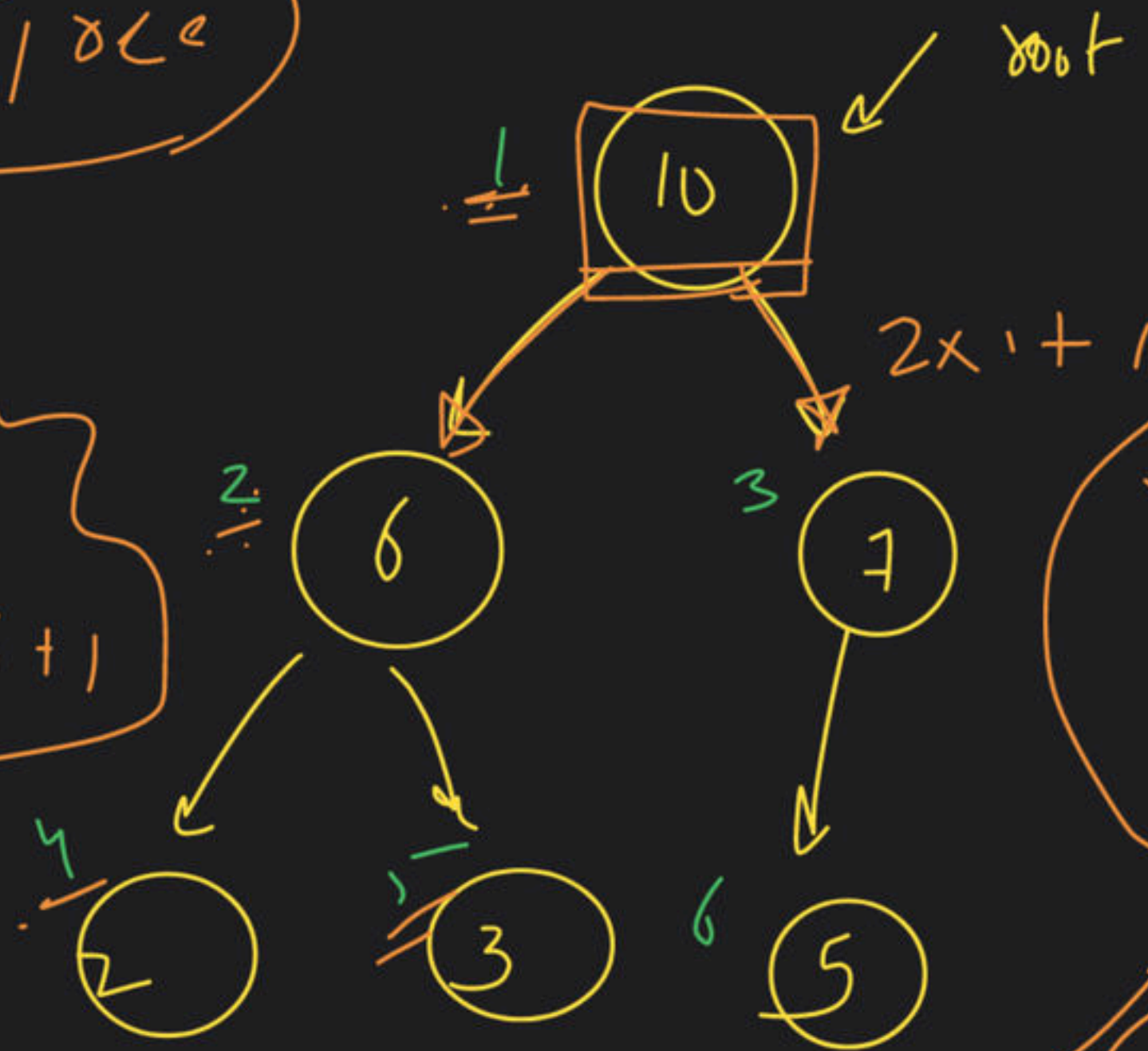
$0 \times 10 \times \text{---}$

→ Heap → visualization → Tree

Impl

array

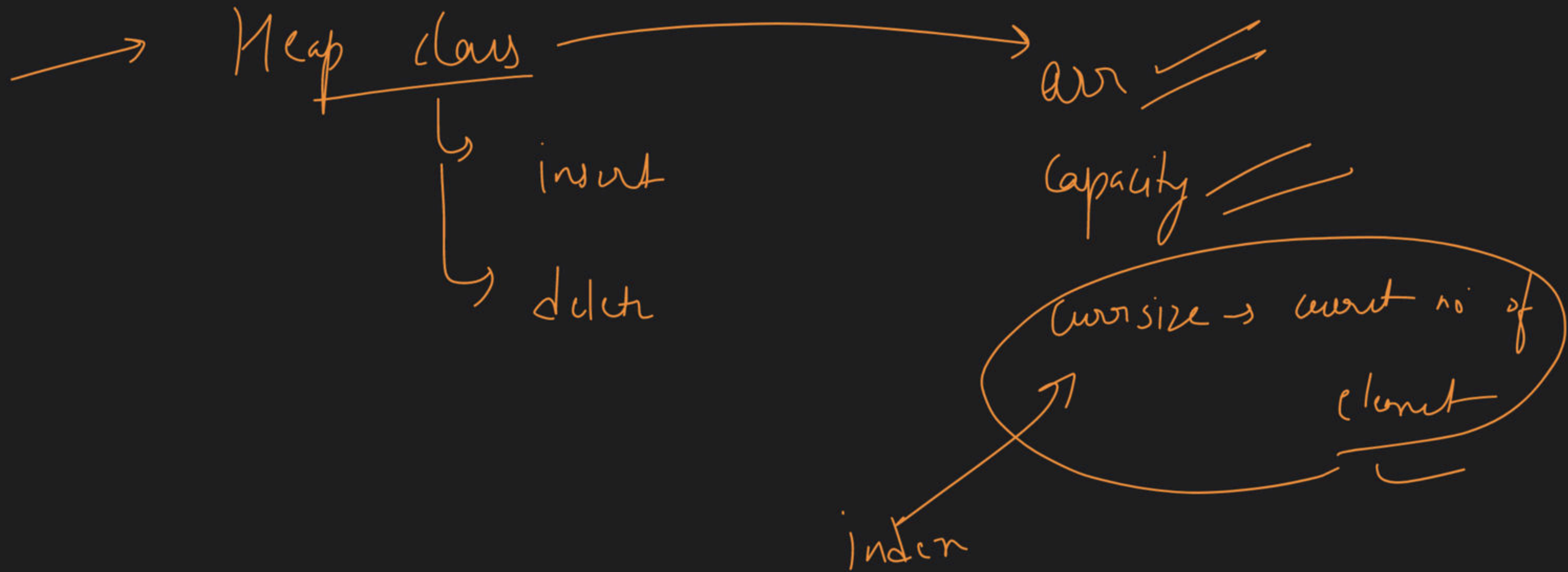
parent = i
left child = $2i$
right child = $2i + 1$



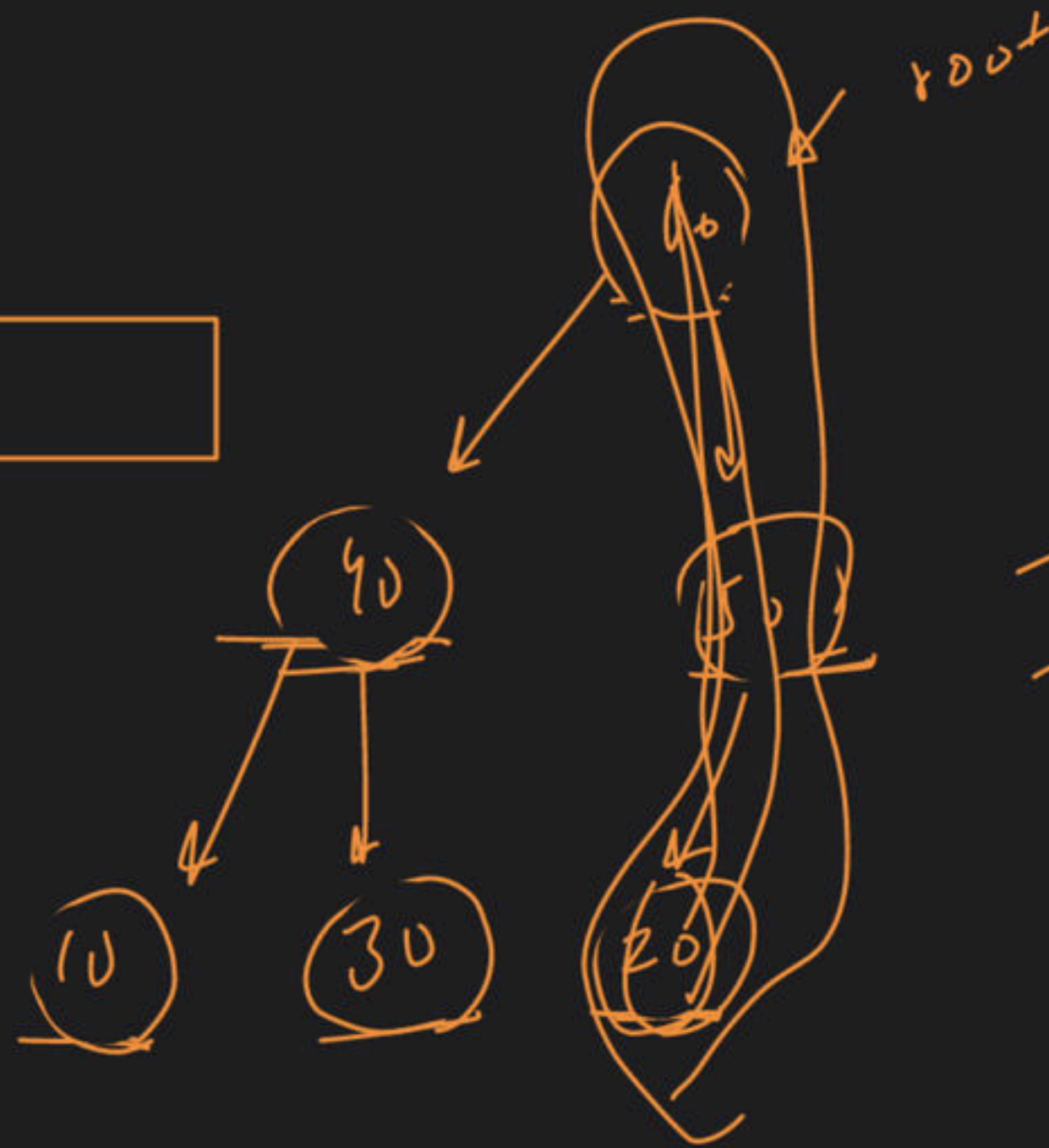
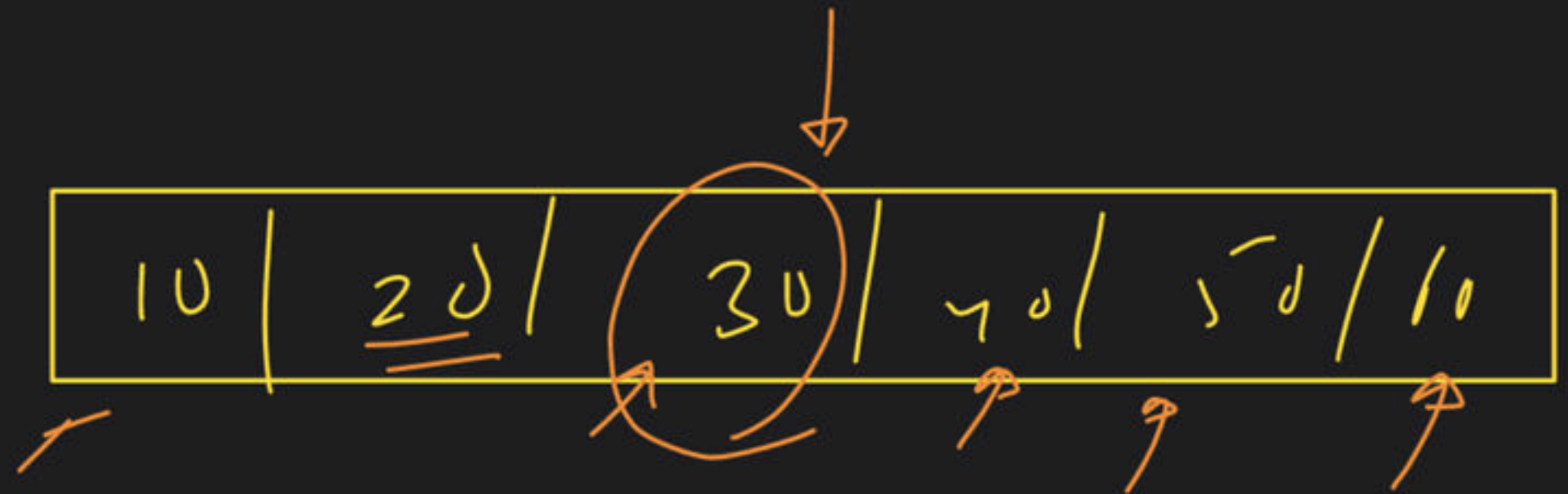
right child = i
↓
parent = $i/2$

left child = i
↓
parent = $i/2$

X	10	6	7	2	3	5
0	1	2	3	4	5	6



→ insert
 ↳ $O(\log n)$



(A) insert → Level order wise

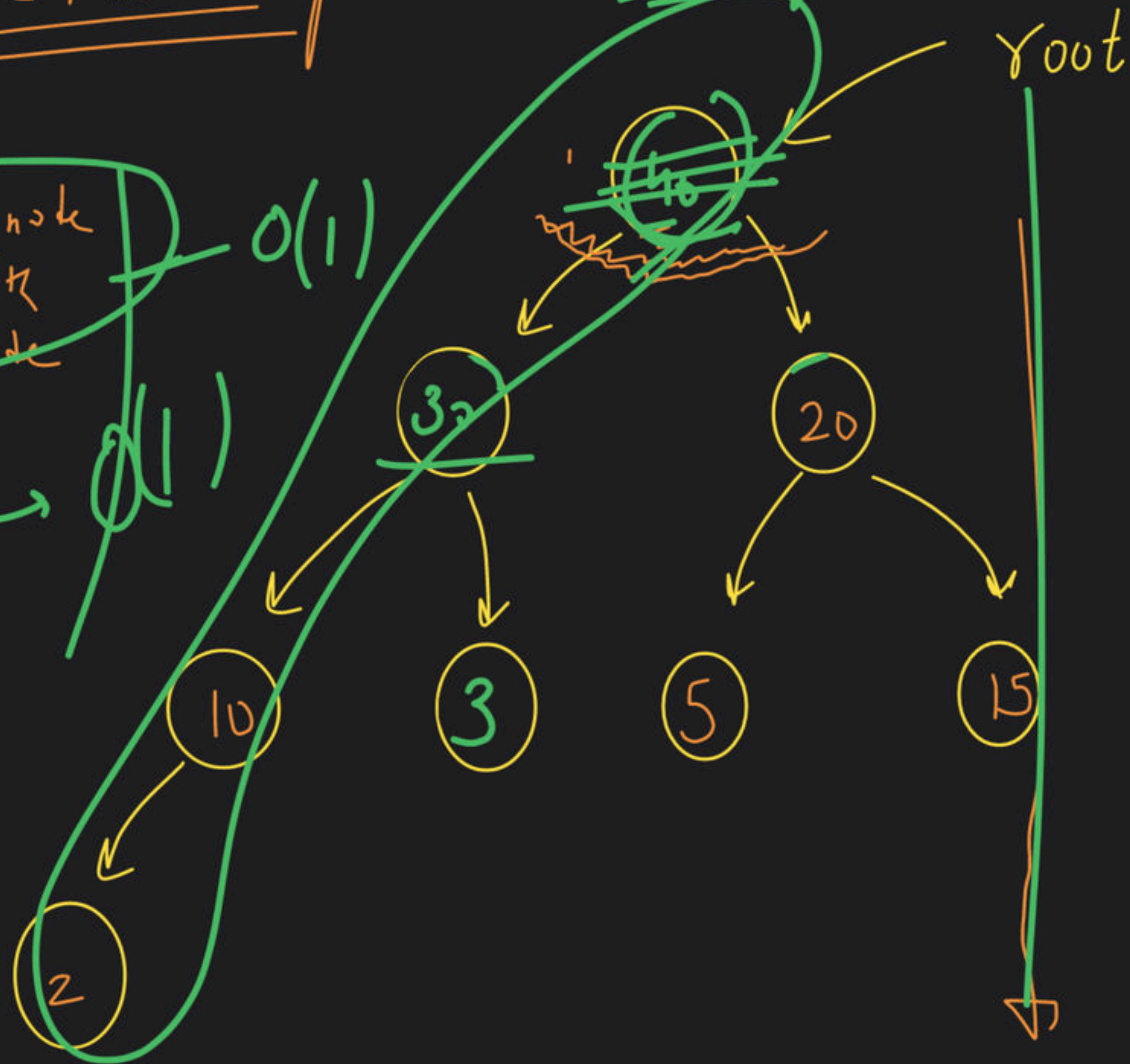
(B) node → right position
 ↓
 by comparing with parent

→ delete from Heap → $O(\log n)$

① replace root node with last node $O(1)$

② $n \rightarrow \text{index}$ $O(1)$

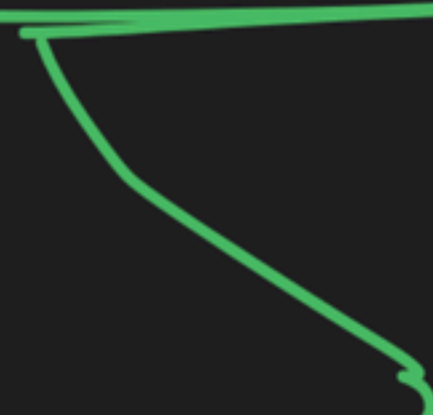
③ arr[i] → val
right position
↓
heapify



heapify
↓
index
left / right
val
array Value Index
↓ ↓
L R
swap (in array)
// Back to recursion

→ insert $\rightarrow O(\log n)$

arr $\rightarrow$ n element

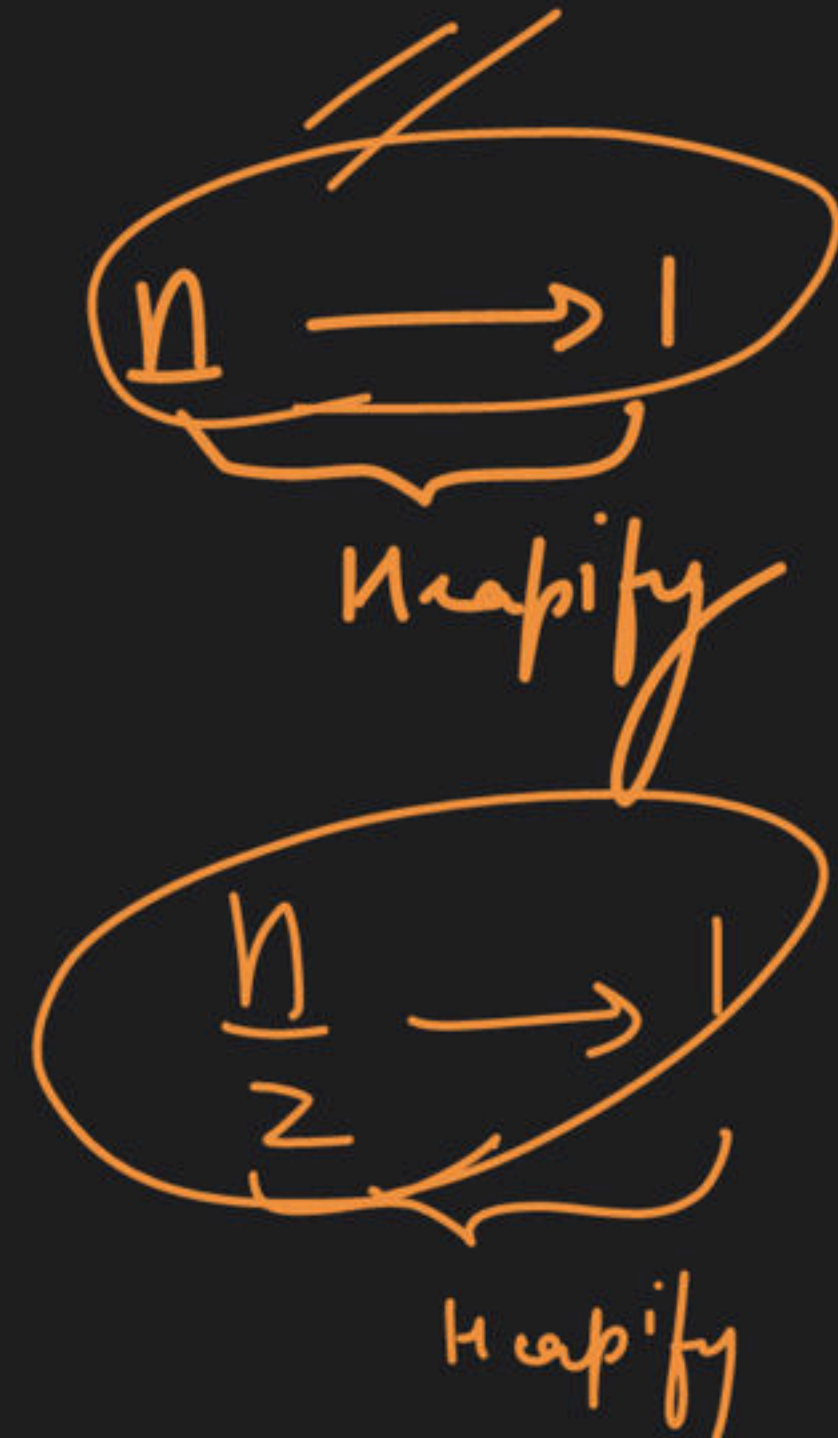
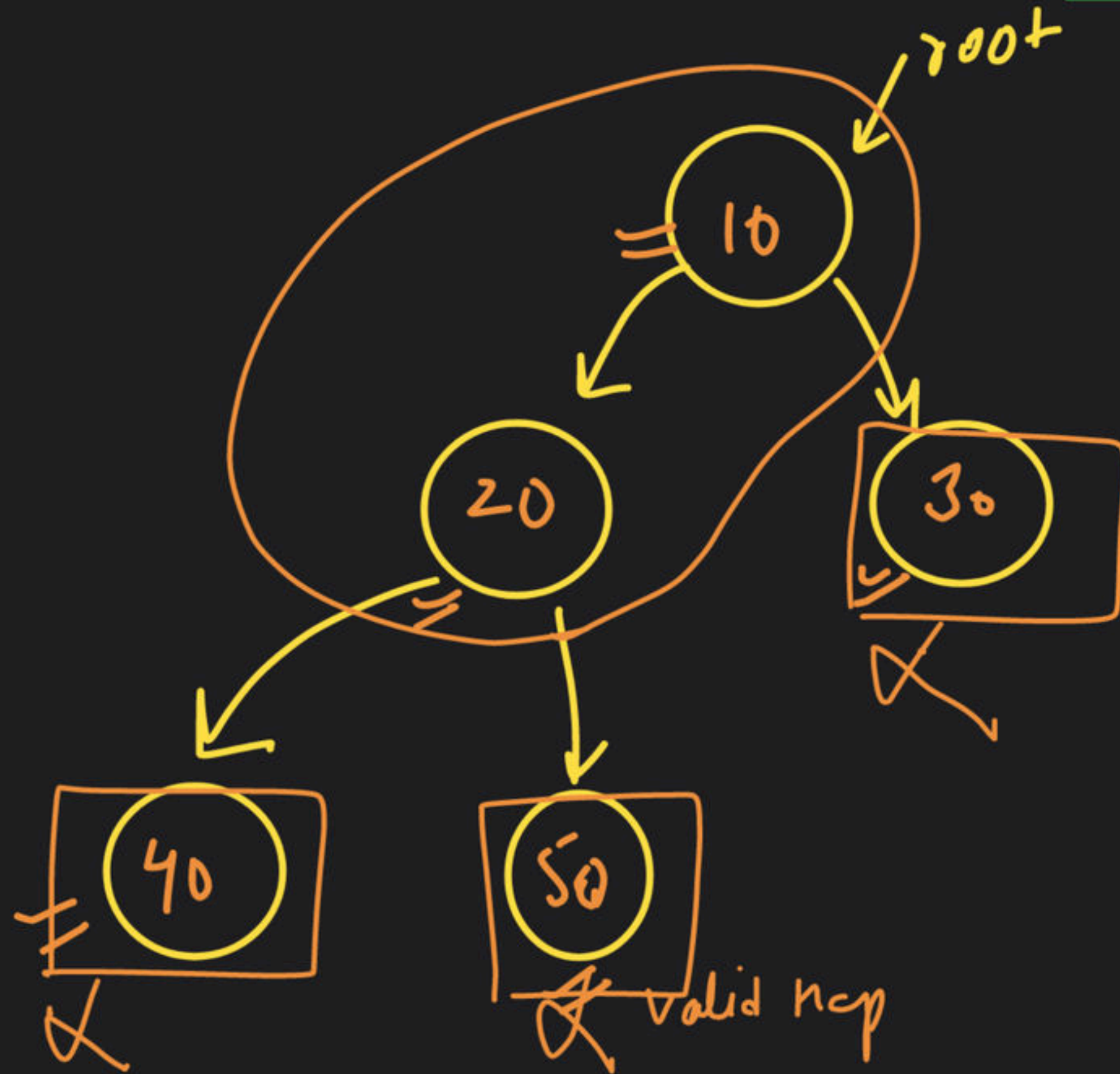


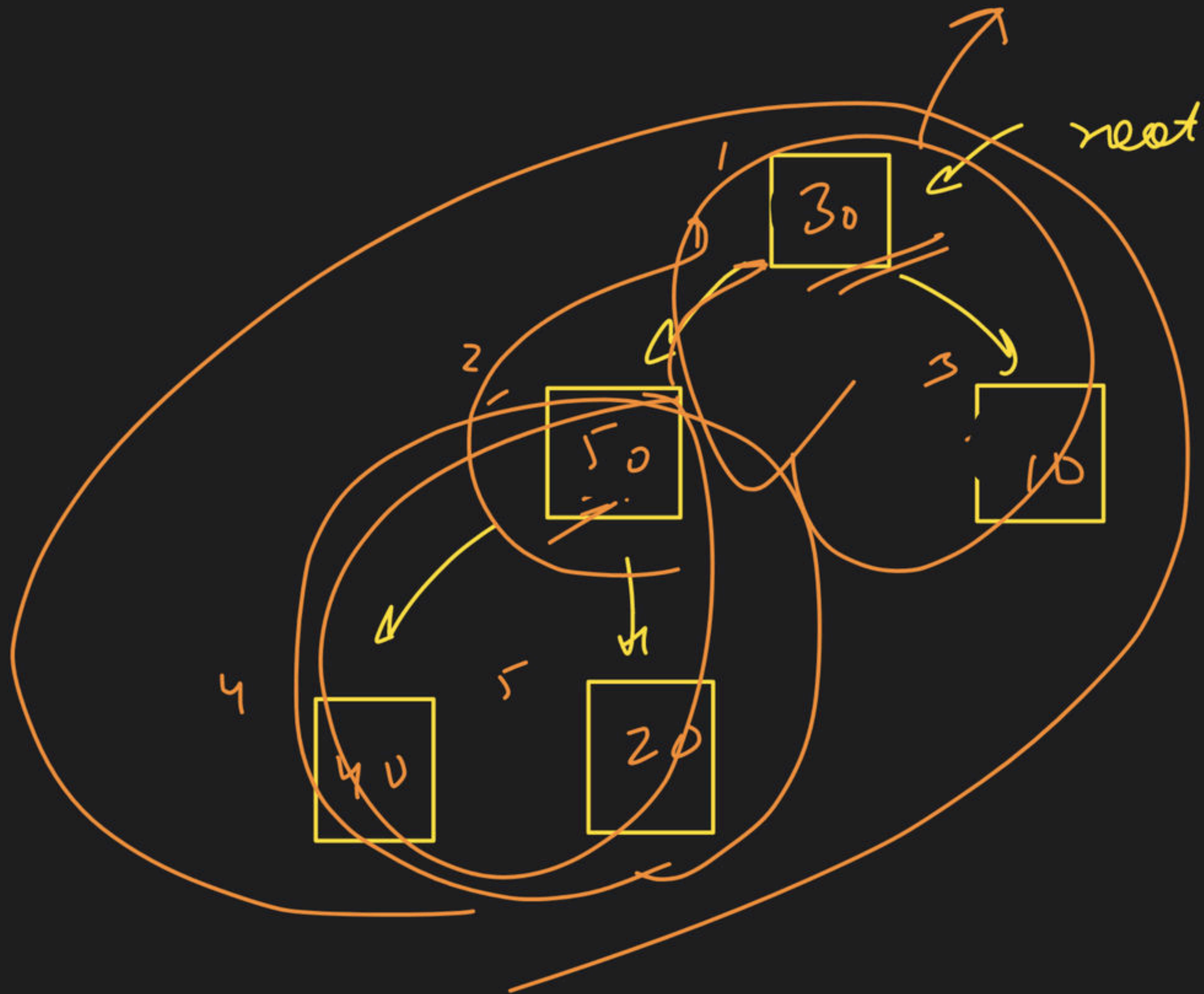
Build Map $\rightarrow$ $\frac{n \log n}{2}$

in-place

$O(n)$

1	2	3	4	5
10	20	30	40	50





$$\frac{n}{2} - 1$$

or

$$1 \rightarrow \frac{n}{2}$$

$$1 \rightarrow \frac{n}{2}$$

$$1 \rightarrow 2$$

Heap Sort:-

1 2 3 4 5
~~50~~, 40, 30, 10, 20
20

(n-1) baar
delux op

~~40~~, 20, 30, 10 } 50
10

after
heapify → ~~30~~, 20, 10 } 40, 50
10

~~20~~, 10 } 30, 40, 50
10

10 } 20, 30, 40, 50 → Sort?





































